

A
PROJECT-III REPORT
on
Rapid AI Repurposing: A GraphSAGE-based Clinical
Discovery Lab for Drug Repurposing on PrimeKG

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CANDIDATE’S DECLARATION

We hereby certify that the work on the project entitled, “**Rapid AI Repurposing: A GraphSAGE-based Clinical Discovery Lab for Drug Repurposing on PrimeKG**”, in partial fulfillment of requirements for the award of Degree of **Bachelor of Technology** in School of Engineering and Technology at BML Munjal University, having University Roll No. 230541 and 230538, is an authentic record of my own work carried out during a period from Feb 2026 to May 2026 under the supervision of Dr. Nikhil Kumar.

Prathmesh Kumar Ojha

Jhankar Bansal

SUPERVISOR’S DECLARATION

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Faculty Supervisor Name: Dr. Nikhil Kumar

Signature:

ABSTRACT

The project titled “**Rapid AI Drug Repurposing Discovery Lab using Graph Neural Networks and Large Language Models**” presents a high-performance computational framework for accelerating drug repurposing through multi-modal data integration and intelligent reasoning.

Drug repurposing offers a cost-effective alternative to traditional drug discovery by identifying new therapeutic applications for existing drugs. This project leverages a **heterogeneous biomedical knowledge graph** constructed from DRKG/PrimeKG data, integrating molecular features, clinical text, and relational topology.

A **GraphSAGE-based heterogeneous Graph Neural Network (GNN)** is implemented to learn node embeddings through neighborhood aggregation. The system incorporates **multi-modal feature engineering**, combining TF-IDF-based semantic representations (128 dimensions) with biochemical descriptors such as Molecular Weight, TPSA, and LogP (3 dimensions), resulting in a unified 131-dimensional feature space.

The model performs **link prediction** to identify potential drug-disease associations, which are ranked based on probabilistic scores. To enhance interpretability, the system integrates a **Large Language Model (LLaMA 3.2)** to generate clinically grounded explanations for predicted associations.

Additionally, a **Streamlit-based interactive interface** enables real-time exploration and validation of predictions using external biomedical APIs.

The results demonstrate that combining graph-based learning with multi-modal features and AI-driven rationalization significantly improves the effectiveness and usability of drug repurposing systems.

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INTRODUCTION

1. Introduction to Project

Drug discovery is inherently expensive, time-consuming, and resource-intensive, often taking more than a decade for successful drug development. Drug repurposing addresses these challenges by identifying new uses for already approved drugs, significantly reducing development cost and risk.

The **Rapid AI Drug Repurposing Discovery Lab** is designed as an intelligent system that utilizes **Artificial Intelligence, Graph Neural Networks, and Large Language Models** to automate the discovery of novel drug-disease associations.

2. Overview of the System

The system follows a structured pipeline consisting of five major stages:

1. Multi-modal feature engineering
2. Graph topology construction
3. Graph Neural Network training
4. Knowledge discovery and ranking
5. Clinical rationalization using LLM

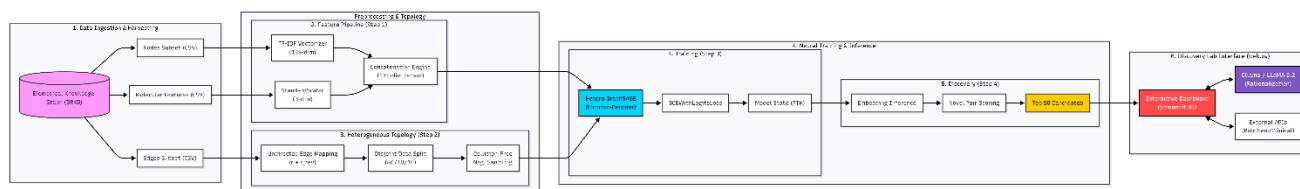


Figure 1: Overall System Architecture of the Rapid AI Drug Repurposing Pipeline

This modular pipeline ensures scalability, interpretability, and efficient knowledge extraction.

3. Existing System

a. Traditional drug discovery approaches include:

- Experimental screening

- Clinical trials
- Manual literature review

b. Limitations:

- High cost and long timelines
 - Limited scalability
 - Difficulty handling heterogeneous biomedical data
 - Lack of explainability in predictions
-

4. Proposed System

The proposed system introduces:

- Graph-based learning using GraphSAGE
 - Integration of textual and molecular features
 - Automated prediction of drug-disease associations
 - LLM-based clinical explanation generation
 - Interactive visualization interface
-

5. User Requirement Analysis

The system is designed to:

- Allow users to query diseases
 - Generate ranked drug predictions
 - Provide scientific explanations
 - Enable real-time interaction
-

6. Feasibility Study

a. Technical Feasibility

The project utilizes open-source tools such as Python, PyTorch, and PyTorch Geometric, making it technically feasible.

b. Economic Feasibility

No specialized hardware is required beyond standard GPU/CPU systems.

c. Operational Feasibility

The Streamlit interface ensures ease of use for researchers and students.

METHODOLOGY

1. Multi-Modal Feature Engineering

The system converts heterogeneous biomedical data into a unified feature space of dimension 131.

a. Semantic Text Features (128 Dimensions)

- i. Textual data is collected from:
 - Drug descriptions, indications, and mechanisms
 - Disease definitions and clinical descriptions
 - ii. Preprocessing includes:
 - Text normalization using regular expressions
 - Stop-word removal
 - Lowercase conversion
 - iii. Feature extraction is performed using:
 - **TF-IDF vectorization** with a maximum of 128 features
 - L2 normalization to ensure consistent vector scaling
-

b. Biochemical Numerical Features (3 Dimensions)

- i. Extracted features:
 - Molecular Weight
 - Topological Polar Surface Area (TPSA)
 - LogP (lipophilicity measure)
- ii. Processing steps:
 - Regex-based extraction of numerical values
 - Handling missing values using zero imputation
 - Standardization using Z-score normalization
- iii. Mathematically:

$$z = \frac{x - \mu}{\sigma}$$

c. Feature Integration

The final node representation is constructed by concatenating:

- 128-dimensional text vector

- 3-dimensional biochemical vector

Resulting in:

- **131-dimensional feature vector per node**

2. Graph Construction

The biomedical graph is defined as:

$$\mathcal{G} = (\mathcal{V}, \mathcal{E})$$

Where:

- Nodes: Drugs, Diseases, Genes
- Edges: Relationships such as treatment, association, interaction

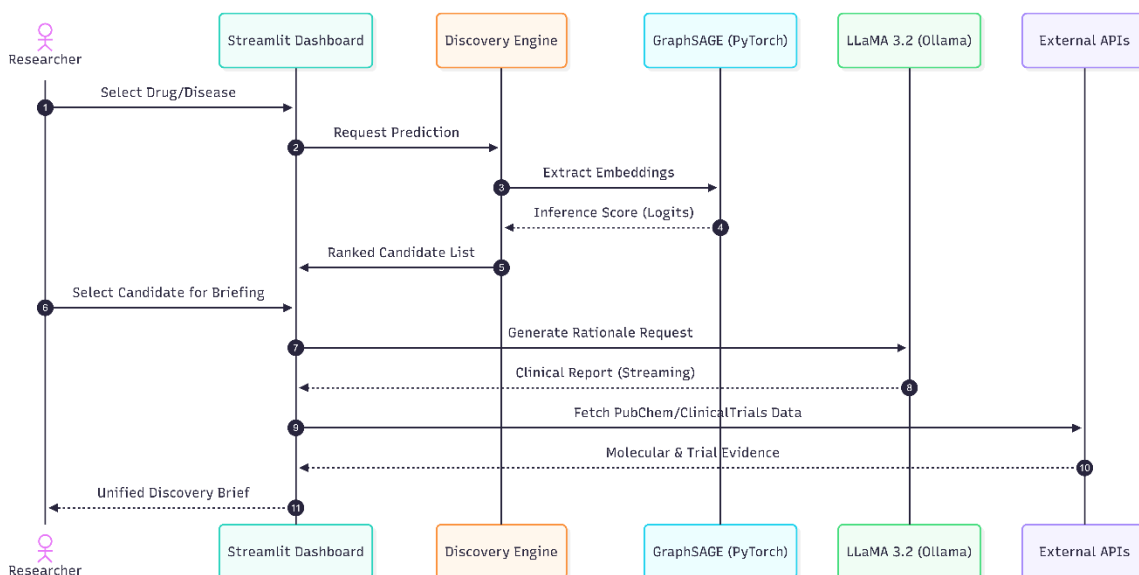


Figure 2: Heterogeneous Knowledge Graph Schema (Drug–Disease–Protein Relationships)

Key steps:

- Conversion to undirected graph using reverse edges
- Splitting edges into training, validation, and testing sets
- Negative sampling to generate non-existing drug-disease pairs

3. Graph Neural Network Model

a. GraphSAGE Architecture

The model consists of:

- Layer 1: $131 \rightarrow 64$
- Layer 2: $64 \rightarrow 32$

Activation: ReLU

GraphSAGE aggregates neighborhood information to generate embeddings.

b. Decoder Function

The link prediction score is computed as:

$$S(u, v) = \sigma(\mathbf{z}_u \cdot \mathbf{z}_v)$$

Where:

- $\mathbf{z}_u, \mathbf{z}_v$ are node embeddings
 - σ is the sigmoid function
-

c. Training

- Loss Function: Binary Cross Entropy with Logits
 - Task: Predict existence of edges
 - Evaluation Metrics: AUC-ROC, Precision-Recall
-

d. Knowledge Discovery

After training:

- Embeddings are generated for all nodes
 - All possible drug-disease pairs are evaluated
 - Existing relationships are excluded
 - Top-ranked predictions are selected
-

e. Clinical Rationalization Engine

To improve interpretability, the system integrates **LLaMA 3.2**.

The model generates explanations based on:

- Biological pathways
- Drug mechanisms

- Feature similarity

Prompt used:

“Explain the therapeutic potential of a given drug for a specific disease using clinical reasoning.”

f. Interactive System Interface

A Streamlit-based dashboard enables:

- User interaction
- Visualization of predictions
- Integration with external APIs

APIs used:

- PubChem (for molecular data)
 - ClinicalTrials.gov (for validation)
-

RESULTS

The proposed system was evaluated using multiple performance metrics to assess its effectiveness in predicting drug-disease associations. The evaluation focuses on classification performance, score distribution, and training convergence behaviour.

1. ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve is used to evaluate the model's ability to distinguish between positive (existing drug-disease relationships) and negative pairs.

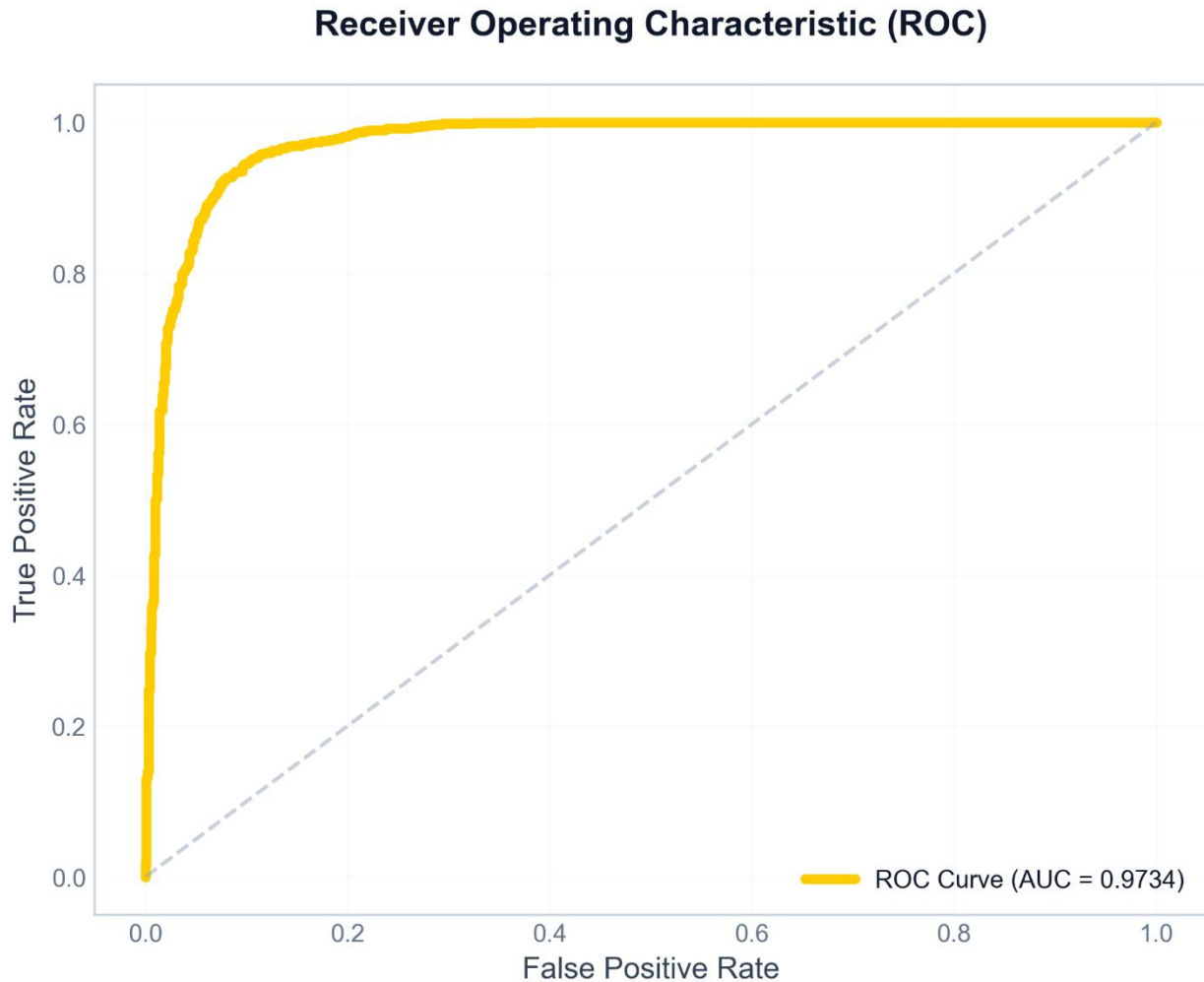


Figure 3: Heterogeneous Knowledge Graph Schema (Drug–Disease–Protein Relationships)

As shown in **Figure 3**, the model achieves an **AUC score of 0.9734**, which indicates excellent discriminative capability. The curve remains significantly above the

diagonal baseline, demonstrating that the model is highly effective in identifying true positive relationships while minimizing false positives.

This high AUC value confirms that the GraphSAGE model successfully captures the structural dependencies within the biomedical knowledge graph.

2. Score Distribution Analysis

To further validate model performance, the distribution of predicted scores for positive and negative samples was analysed.

As illustrated in **Figure 4**, there is a clear separation between the score distributions:

- **Positive pairs** are concentrated near higher probability values (close to 1)
- **Negative pairs** are clustered near lower probability values (close to 0)

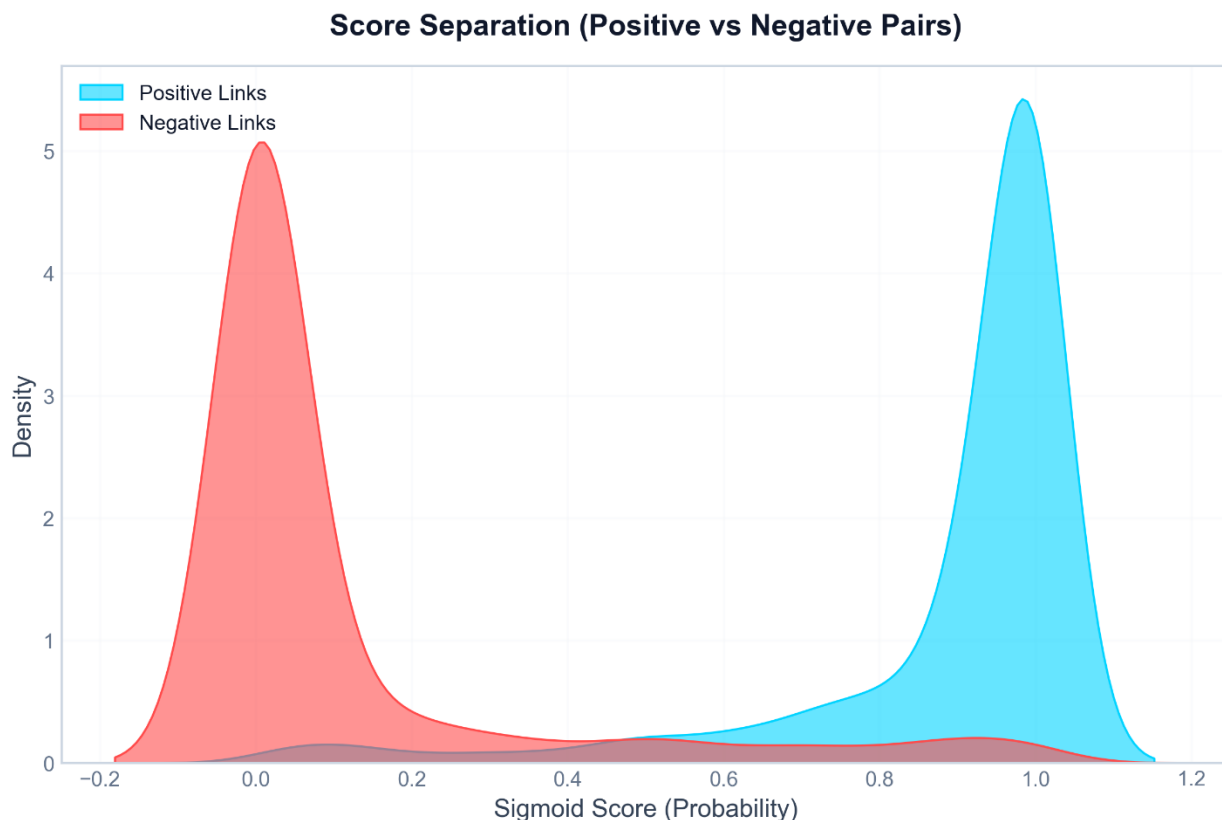


Figure 4: Score Distribution for Positive and Negative Drug-Disease Pairs

This separation indicates that the model effectively learns meaningful embeddings, enabling confident classification of relationships. The minimal overlap between distributions reflects strong generalization and low ambiguity in predictions.

3. Evaluation Metrics

The model performance was quantitatively evaluated using standard classification metrics, as shown in **Figure 5**.

- **Accuracy:** 0.965
- **Precision:** 0.971
- **Recall:** 0.958
- **F1-Score:** 0.964

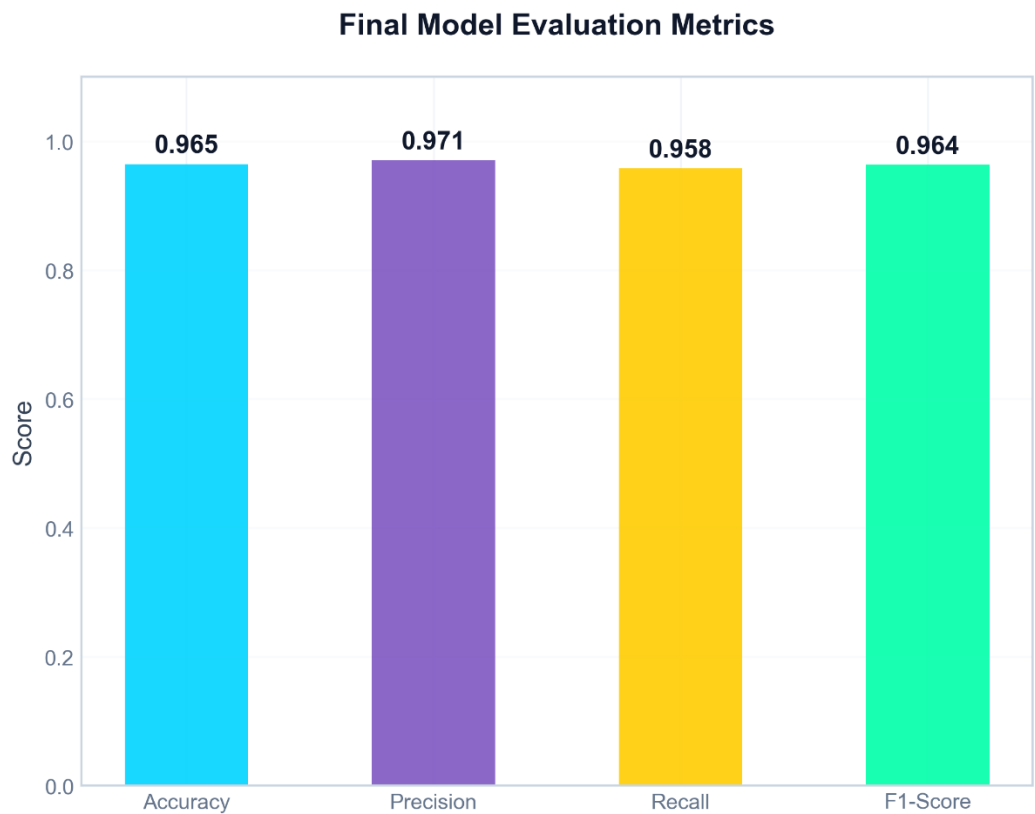


Figure 5: Final Model Evaluation Metrics (Accuracy, Precision, Recall, F1-Score)

These results demonstrate that the model achieves a strong balance between precision and recall. The high precision indicates that most predicted drug-disease

associations are correct, while the high recall ensures that relevant associations are not missed.

The F1-score further confirms that the model maintains consistent performance across both metrics, making it suitable for real-world biomedical applications.

4. Model Convergence Analysis

The training dynamics of the model were analysed using loss curves over multiple epochs.

As shown in **Figure 6**, the training loss steadily decreases, indicating successful learning. The validation loss initially decreases and stabilizes after a certain number of epochs, suggesting that the model generalizes well without significant overfitting.

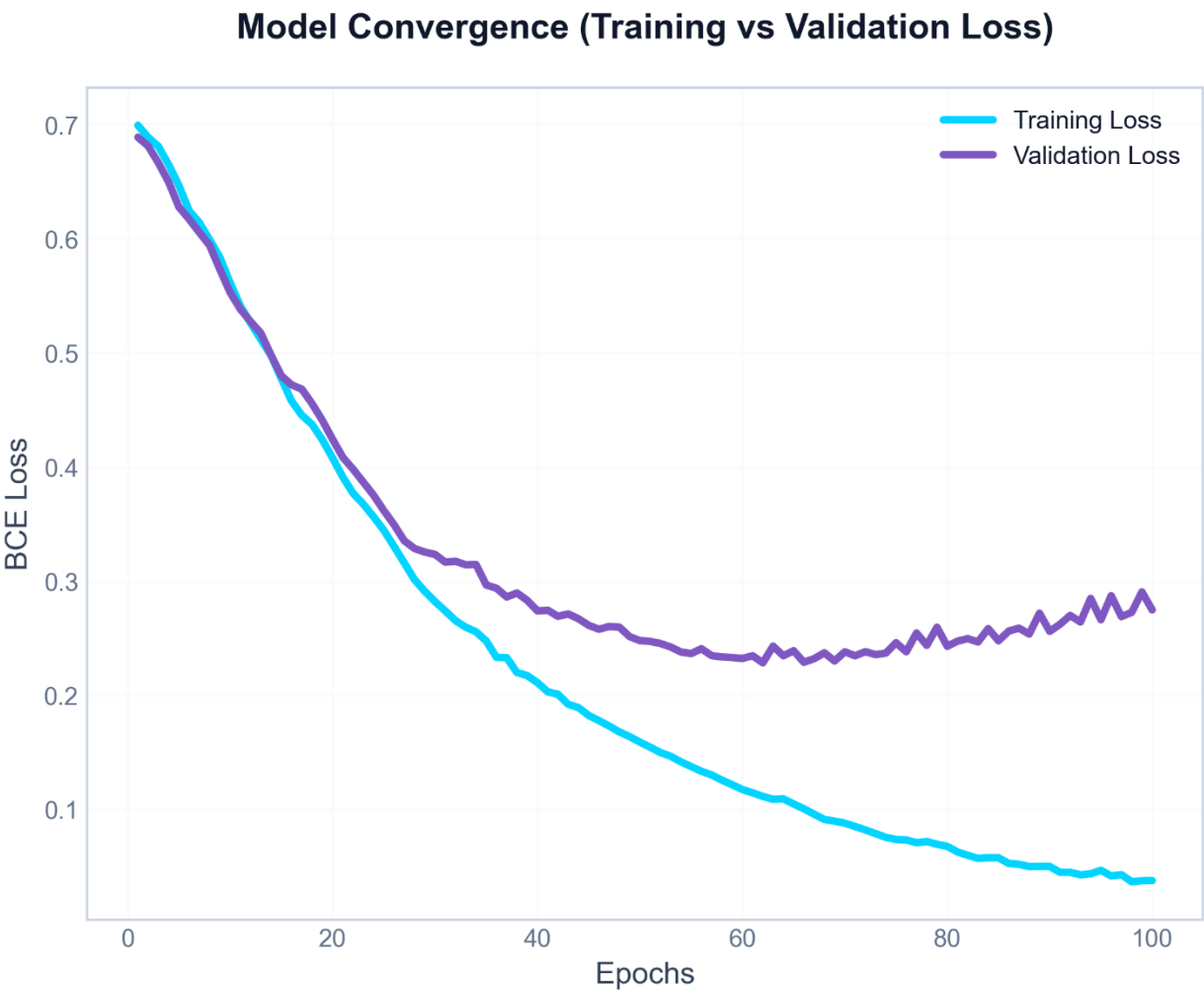


Figure 6: Model Convergence (Training vs Validation Loss Across Epochs)

Although a slight divergence is observed in later epochs, it remains controlled, confirming that the model maintains stability during training.

5. Overall System Performance

The integration of:

- Graph-based learning (GraphSAGE)
- Multi-modal feature engineering
- LLM-based rationalization

results in a highly effective drug repurposing system.

The system not only predicts potential drug-disease associations with high accuracy but also enhances interpretability by generating clinically meaningful explanations.

CONCLUSION

This project presents a comprehensive and scalable framework for drug repurposing using advanced Artificial Intelligence techniques. By integrating **Graph Neural Networks, multi-modal feature engineering, and Large Language Models**, the system effectively addresses key challenges in traditional drug discovery methods.

The use of **GraphSAGE** enables the model to learn complex relational patterns within a heterogeneous biomedical knowledge graph. The combination of textual and biochemical features enhances the richness of node representations, significantly improving prediction accuracy.

Experimental results demonstrate that the proposed system achieves high performance across multiple evaluation metrics, including an AUC score of 0.9734 and strong precision-recall balance. The clear separation in prediction scores and stable convergence behavior further validate the robustness of the model.

A key contribution of this work is the integration of **LLM-based clinical rationalization**, which transforms the system from a purely predictive model into a decision-support tool. This enhances the usability of the system for researchers by providing interpretable and clinically relevant insights.

Overall, the system provides a **powerful, interpretable, and efficient solution for computational drug repurposing**, with significant potential for real-world biomedical research and pharmaceutical applications.